2.4 Field Homomorphisms

Section 2.3 looked at fields inside fields. This section looks at maps between fields. Two such maps have appeared already. Complex conjugation $z \mapsto \overline{z}$ turns sums into sums and products into products (Theorem 1.2.2 (a), (b)), and so does $a + b\sqrt{2} \mapsto a - b\sqrt{2}$ on $\mathbb{Q}(\sqrt{2})$ (Number Systems: From \(\nN\) to \(\nR\), Example (Conjugation in $\mathbb{Q}(\sqrt{2})$)). A map that respects both operations carries every equation that holds in one field to an equation in the other. That single observation decides where such a map can send an element, shows that it never collapses two elements, and rules out most maps between most pairs of fields.

Throughout, F and L are fields, and $n \cdot a$ for $n \in \mathbb{N}$ is the multiple of *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Section 2.3 ("Multiples"): $0 \cdot a \coloneqq 0$, $(n+1) \cdot a \coloneqq n \cdot a + a$. Powers in a field are defined by the same recursion with products: $a^0 \coloneqq 1$ and $a^{n+1} \coloneqq a^n a$ for $n \in \mathbb{N}$, as Sets, Relations, and Cardinality, Definition 2.3.2 does for real numbers; the power laws $a^{m+n} = a^m a^n$, $(a^m)^n = a^{mn}$ and $(ab)^n = a^n b^n$ of Sets, Relations, and Cardinality, Proposition 2.3.4 have proofs that use only (F1)–(F7), (F9) and induction, and hold in every field with the same proofs.

2.4.1 Maps that respect both operations

A homomorphism is a map that respects addition, multiplication and the number one.

Definition 2.4.1. Field Homomorphism

Let F and L be fields. A **homomorphism** (of fields) $\varphi\colon F \to L$ is a function such that, **for all** $a, b \in F$,

$$\varphi(a+b) = \varphi(a) + \varphi(b), \qquad \varphi(ab) = \varphi(a)\varphi(b), \qquad \varphi(1) = 1.$$

A **bijective** (Sets, Relations, and Cardinality, Definition 1.4.5) homomorphism is an **isomorphism**; F and L are **isomorphic**, written $F \cong L$, if there is an isomorphism $F \to L$. An isomorphism $F \to F$ is an **automorphism** of F.

In words: the first clause says that adding and then mapping gives the same as mapping and then adding; the second says the same for multiplying; the third pins down the image of 1, where 1 on the left is 1_F and on the right 1_L. An isomorphism is a homomorphism that is also a one-to-one correspondence of elements, so F and L are then "the same field with the elements renamed".

Examples.

- The **identity** $\mathrm{id}_F\colon F \to F$ is an automorphism.
- **Conjugation** $z \mapsto \overline{z}$ on $\mathbb{C}$ is a homomorphism by Theorem 1.2.2 (a), (b), since $\overline{1} = 1$; Example (Conjugation and Other Automorphisms) below shows that it is an automorphism.
- **The conjugation of** $\mathbb{Q}(\sqrt{2})$, $\sigma(a + b\sqrt{2}) \coloneqq a - b\sqrt{2}$, is an automorphism (Number Systems: From \(\nN\) to \(\nR\), Example (Conjugation in $\mathbb{Q}(\sqrt{2})$)).

Why the clause $\varphi(1) = 1$. The **zero map** $a \mapsto 0$ satisfies the first two clauses, since $0 = 0 + 0$ and $0 = 0 \cdot 0$, and it carries no information about F at all. The third clause excludes it, and the next proposition shows that the third clause is exactly what forces a homomorphism to remember every element.

Remark. The order-preserving field maps of *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Section 6.1 (Number Systems: From \(\nN\) to \(\nR\), Definition 6.1.1), are the homomorphisms between ordered fields that also preserve $<$. From here on, $F \cong L$ means isomorphic as fields, in the sense above; the relation of that chapter, an isomorphism that also preserves the order, is said in words, "isomorphic as ordered fields". The two differ: the conjugation σ of $\mathbb{Q}(\sqrt{2})$ is an automorphism of the field, and it is not order-preserving, since $0 < \sqrt{2}$ while $\sigma(\sqrt{2}) = -\sqrt{2} < 0 = \sigma(0)$.

2.4.2 What a homomorphism must do

To find where a homomorphism sends an element, we take an equation that the element satisfies and apply φ to both sides. The equation in L that comes out often has only one solution.

Proposition 2.4.1. Properties of Field Homomorphisms

Let $\varphi\colon F \to L$ be a homomorphism of fields, and let $a, b \in F$.

(a) $\varphi(0) = 0$, $\varphi(-a) = -\varphi(a)$ and $\varphi(a - b) = \varphi(a) - \varphi(b)$.

(b) If $a \neq 0$, then $\varphi(a) \neq 0$ and $\varphi(a^{-1}) = \varphi(a)^{-1}$.

(c) φ is **injective**.

(d) The image $\varphi(F)$ is a subfield of L, and φ, with codomain $\varphi(F)$, is an isomorphism $F \to \varphi(F)$.

(e) If $\psi\colon L \to M$ is a homomorphism of fields, then $\psi \circ \varphi\colon F \to M$ is a homomorphism. If φ is an isomorphism, its inverse function $\varphi^{-1}\colon L \to F$ is an isomorphism. The identity id_F is an automorphism.

(f) $\varphi(n \cdot a) = n \cdot \varphi(a)$ and $\varphi(a^n) = \varphi(a)^n$ for every $n \in \mathbb{N}$.

Idea. For (a) and (b) the equations to push through φ are $0+0 = 0$, $a+(-a) = 0$ and $aa^{-1} = 1$; in L the results are equations with a unique solution, by cancellation and the uniqueness of negatives and inverses. For (c): a homomorphism cannot send a non-zero element to 0, because a non-zero element has an inverse, and then $\varphi(a)\varphi(a^{-1}) = \varphi(1) = 1 \neq 0$. Injectivity is that fact applied to $a - b$. The rest is bookkeeping with (a)–(c), and induction for (f).

Before reading on: in the proof of (c), where is $\varphi(1) = 1$ *used, and which map satisfies the other two clauses of Definition 2.4.1 but is not injective?*

Proof.

(a) Since $0 + 0 = 0$ in F,

$$\varphi(0) + \varphi(0) = \varphi(0 + 0) = \varphi(0) = \varphi(0) + 0,$$

and additive cancellation in L (Theorem 2.1.1 (b)) gives $\varphi(0) = 0$. Next, $a + (-a) = 0$ gives $\varphi(a) + \varphi(-a) = \varphi(a + (-a)) = \varphi(0) = 0$, so $\varphi(-a)$ is a negative of $\varphi(a)$, and $\varphi(-a) = -\varphi(a)$ by the uniqueness of negatives (Theorem 2.1.1 (c)). Finally, $\varphi(a - b) = \varphi(a + (-b)) = \varphi(a) + \varphi(-b) = \varphi(a) - \varphi(b)$.

(b) Let $a \neq 0$. Since $aa^{-1} = 1$,

$$\varphi(a)\varphi(a^{-1}) = \varphi(aa^{-1}) = \varphi(1) = 1.$$

If $\varphi(a) = 0$, the left side would be $0 \cdot \varphi(a^{-1}) = 0$ (Theorem 2.1.1 (d)), and the display would read $0 = 1$, against (F7) in L. So $\varphi(a) \neq 0$, and the display says that $\varphi(a^{-1})$ is an inverse of $\varphi(a)$; by the uniqueness of inverses (Theorem 2.1.1 (c)), $\varphi(a^{-1}) = \varphi(a)^{-1}$.

(c) Let $\varphi(a) = \varphi(b)$. By (a), $\varphi(a - b) = \varphi(a) - \varphi(b) = 0$. By (b), read contrapositively, an element whose image is 0 is 0; so $a - b = 0$, and $a = b$ (add b to both sides, with (F2), (F1), (F4) and (F3)). So φ is injective.

(d) We use Proposition 2.3.1 (c) for $\varphi(F) \subseteq L$. First, $1 = \varphi(1) \in \varphi(F)$. Next, two elements of $\varphi(F)$ are $\varphi(a)$, $\varphi(b)$ with $a, b \in F$ (Sets, Relations, and Cardinality, Definition 1.6.1), and $\varphi(a) - \varphi(b) = \varphi(a - b) \in \varphi(F)$ by (a). If moreover $\varphi(b) \neq 0$, then $b \neq 0$, since $\varphi(0) = 0$; so by (b), $\varphi(a)\varphi(b)^{-1} = \varphi(a)\varphi(b^{-1}) = \varphi(ab^{-1}) \in \varphi(F)$. Hence $\varphi(F)$ is a subfield of L, a field with the operations of L (Proposition 2.3.1 (a)). The map $\varphi\colon F \to \varphi(F)$ still satisfies the three clauses, since the operations of $\varphi(F)$ are those of L and its one is 1_L; it is surjective by the definition of the image, and injective by (c). So it is an isomorphism.

(e) For the composite, $(\psi \circ \varphi)(a + b) = \psi(\varphi(a) + \varphi(b)) = \psi(\varphi(a)) + \psi(\varphi(b))$, likewise for products, and $\psi(\varphi(1)) = \psi(1) = 1$ (Sets, Relations, and Cardinality, Definition 1.4.2). Now let φ be an isomorphism, with inverse function $\varphi^{-1}\colon L \to F$ (Sets, Relations, and Cardinality, Definition 1.5.2, which exists by Sets, Relations, and Cardinality, Theorem 1.5.4). For $x, y \in L$,

$$\varphi(\varphi^{-1}(x) + \varphi^{-1}(y)) = \varphi(\varphi^{-1}(x)) + \varphi(\varphi^{-1}(y)) = x + y = \varphi(\varphi^{-1}(x + y)),$$

so $\varphi^{-1}(x) + \varphi^{-1}(y) = \varphi^{-1}(x + y)$, because φ is injective. The same computation with products gives $\varphi^{-1}(x)\varphi^{-1}(y) = \varphi^{-1}(xy)$. And $\varphi(1) = 1$ gives $\varphi^{-1}(1) = 1$. The function φ^{-1} is bijective, with inverse φ (Sets, Relations, and Cardinality, Theorem 1.5.4); so it is an isomorphism. The identity satisfies the three clauses trivially and is bijective.

(f) By induction on n (Logic and Proof, Axiom 3.2.1, read on $\mathbb{N} = \omega$ through Number Systems: From \(\nN\) to \(\nR\), Corollary 1.7.3). For $n = 0$: $\varphi(0 \cdot a) = \varphi(0) = 0 = 0 \cdot \varphi(a)$ by (a), and $\varphi(a^0) = \varphi(1) = 1 = \varphi(a)^0$. If the two formulas hold for n, then

$$\varphi((n + 1) \cdot a) = \varphi(n \cdot a + a) = n \cdot \varphi(a) + \varphi(a) = (n + 1) \cdot \varphi(a),$$
$$\varphi(a^{n+1}) = \varphi(a^n a) = \varphi(a)^n \varphi(a) = \varphi(a)^{n+1}.$$

This proves the proposition.

■

The answer to the prompt. In (c), $\varphi(1) = 1$ enters through (b): the display $\varphi(a)\varphi(a^{-1}) = 1$ is what makes $\varphi(a) = 0$ impossible. The zero map $F \to L$ satisfies the two other clauses, sends every element to 0, and is not injective. A map that respected sums and products but sent 1 to 0 would be the zero map: $\varphi(a) = \varphi(a \cdot 1) = \varphi(a)\varphi(1) = \varphi(a) \cdot 0 = 0$, by (F7), the product clause, $\varphi(1) = 0$, and Theorem 2.1.1 (d) with (F6) in L.

We call the move of this proof **push the equation through** φ: to learn $\varphi(x)$, apply φ to an equation that x satisfies, and solve the resulting equation in L. By (c) and (d), a homomorphism $F \to L$ is the same thing as a copy of F inside L: $F \cong \varphi(F)$, a subfield of L. By (e) and Sets, Relations, and Cardinality, Theorem 1.4.4 (c), composing isomorphisms gives isomorphisms, so

$\cong$ is reflexive, symmetric and transitive on any set of fields.

Warning. A bijection that respects addition alone is not an isomorphism. The map $f\colon \mathbb{Q} \to \mathbb{Q}$, $f(x) \coloneqq 2x$, is bijective (its inverse is $x \mapsto x/2$) and satisfies $f(a+b) = f(a) + f(b)$, by (F9). But $f(1 \cdot 1) = 2$, while $f(1)f(1) = 4$; and $f(1) = 2 \neq 1$. All three clauses of Definition 2.4.1 must be checked.

Quick check. Is $a \mapsto -a$ a homomorphism $\mathbb{Q} \to \mathbb{Q}$?

Solution. No. It sends 1 to -1, and $-1 \neq 1$ in $\mathbb{Q}$: if $-1 = 1$, then $1 + 1 = 1 + (-1) = 0$, while $1 + 1 = 2 \cdot 1 > 0$ in the ordered field $\mathbb{Q}$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (h), Number Systems: From \(\nN\) to \(\nR\), Theorem 2.5.1). (In $\mathbb{F}_2$, where $-1 = 1$, the same map is the identity.) It also fails the product clause: $1 \cdot 1 = 1$ is sent to -1, while the product of the images is $(-1)(-1) = 1$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (c)). It does respect sums, by Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (c) and (F9): $-(a+b) = (-1)(a+b) = -a + (-b)$.

2.4.3 Examples, and pairs of fields with no map between them

Example. Conjugation and Other Automorphisms

(i) Show that conjugation $z \mapsto \overline{z}$ is an automorphism of $\mathbb{C}$, that it is not the identity, and that $\overline{z} = z$ if and only if $z \in \mathbb{R}$.

(ii) Let K be a subfield of L. Show that the inclusion $K \to L$, $x \mapsto x$, is a homomorphism, and that it is not surjective when $K \neq L$.

(iii) Recall why $a + b\sqrt{2} \mapsto a - b\sqrt{2}$ is an automorphism of $\mathbb{Q}(\sqrt{2})$.

(iv) Which automorphisms does $\mathbb{R}$ have?

Solution.

(i) By Theorem 1.2.2 (a), (b), $\overline{z+w} = \overline{z} + \overline{w}$ and $\overline{zw} = \overline{z}\,\overline{w}$; and $\overline{1} = \overline{1 + 0i} = 1 - 0i = 1$ (Definition 1.2.1). So conjugation is a homomorphism. By Theorem 1.2.2 (a), $\overline{\overline{z}} = z$, so conjugation is its own inverse, hence bijective (Sets, Relations, and Cardinality, Theorem 1.5.4): an automorphism. It is not the identity: $\overline{i} = -i$, and $-i \neq i$, since $i - (-i) = 2i = (0, 2) \neq (0, 0)$.
For the fixed elements, let $z = a + bi$ with $a, b \in \mathbb{R}$. Then $\overline{z} = a - bi$ (Definition 1.2.1), and comparing parts (Definition 1.1.1), $\overline{z} = z$ if and only if $-b = b$, that is $2b = 0$, that is $b = 0$ (as $2 = 2 \cdot 1 > 0$ in the ordered field $\mathbb{R}$, Number Systems: From \(\nN\) to \(\nR\), Theorem 5.3.3, Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (h), so $2 \neq 0$, and by Theorem 2.1.1 (f)). And $b = 0$ says $z = a + 0i = a \in \mathbb{R}$ (Proposition 1.1.1). This fact is not among the parts of Theorem 1.2.2; the two lines above prove it.

(ii) The operations of K are those of L, and $1_K = 1_L$ (Proposition 2.3.1 (a)). So $x \mapsto x$ satisfies the three clauses. If $K \neq L$, some element of L is not in K, and it is not a value of the inclusion.

(iii) This is Number Systems: From \(\nN\) to \(\nR\), Example (Conjugation in $\mathbb{Q}(\sqrt{2})$), recalled and not proved again: unique representation makes the formula a function, the product formula of Example (The Field Generated by the Square Root of Two) with $-b$, $-d$ in place of b, d shows that products go to products, and the map is its own inverse.

(iv) Only the identity. *Number Systems: From $\mathbb{N}$ to $\mathbb{R}$*, Chapter 6, proved that a map $\sigma\colon \mathbb{R} \to \mathbb{R}$ with $\sigma(a+b) = \sigma(a) + \sigma(b)$, $\sigma(ab) = \sigma(a)\sigma(b)$ and $\sigma(1) = 1$, that is, a homomorphism $\mathbb{R} \to \mathbb{R}$, is the identity, with no order condition assumed (Number Systems: From \(\nN\) to \(\nR\), Theorem 6.3.4, with $\mathbb{R}$ complete by Number Systems: From \(\nN\) to \(\nR\), Theorem 5.6.3). So $\mathbb{C}$ has a symmetry and $\mathbb{R}$ has none. Section 2.5 shows that $\mathbb{Q}$ and every $\mathbb{F}_p$ have no automorphism other than the identity either.

Pushing an equation through φ also proves that no homomorphism exists: an equation that holds in F and has no counterpart in L leaves φ nowhere to go.

Example. Pairs of Fields with No Homomorphism

Show that there is no homomorphism (i) $\mathbb{C} \to \mathbb{R}$; (ii) $\mathbb{F}_2 \to \mathbb{Q}$; (iii) $\mathbb{Q} \to \mathbb{F}_p$, for any prime p. (iv) Which clause of Definition 2.4.1 does the zero map $\mathbb{Q} \to \mathbb{Q}$ fail?

Solution.

(i) Suppose $\varphi\colon \mathbb{C} \to \mathbb{R}$ is a homomorphism. In $\mathbb{C}$, $i^2 = -1$. Pushing this through φ,

$$\varphi(i)^2 = \varphi(i^2) = \varphi(-1) = -\varphi(1) = -1,$$

by (f) and (a) of Proposition 2.4.1. But $\mathbb{R}$ is an ordered field (Number Systems: From \(\nN\) to \(\nR\), Theorem 5.3.3), in which every square is ≥ 0 and $1 > 0$, so $-1 < 0$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (e), (a)). So $\varphi(i)^2 = -1$ is impossible.

(ii) Suppose $\varphi\colon \mathbb{F}_2 \to \mathbb{Q}$ is a homomorphism. In $\mathbb{F}_2$, $1 + 1 = 0$ (Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$)). Pushing it through φ, $1 + 1 = \varphi(1) + \varphi(1) = \varphi(1 + 1) = \varphi(0) = 0$ in $\mathbb{Q}$, by (a). But $1 + 1 = 2 \cdot 1 > 0$ in the ordered field $\mathbb{Q}$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (h), Number Systems: From \(\nN\) to \(\nR\), Theorem 2.5.1). So no such φ exists.

(iii) Suppose $\varphi\colon \mathbb{Q} \to \mathbb{F}_p$ is a homomorphism. In $\mathbb{F}_p$, $p \cdot 1 = 0$ (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.3 (d)). By Proposition 2.4.1 (f), $\varphi(p \cdot 1) = p \cdot \varphi(1) = p \cdot 1 = 0 = \varphi(0)$, and φ is injective by (c), so $p \cdot 1 = 0$ in $\mathbb{Q}$. But $p \cdot 1 > 0$ in $\mathbb{Q}$, as in (ii), since $p \geq 1$. So no such φ exists.

(iv) The zero map $a \mapsto 0$ respects sums and products, since $0 + 0 = 0$ and $0 \cdot 0 = 0$, and it sends 1 to $0 \neq 1$. So it fails only $\varphi(1) = 1$, and is not a homomorphism.

Section 2.5 explains (ii) and (iii) at once: both pairs of fields differ in the number of copies of 1 that add up to 0.

2.4.4 Exercises

2.4.4.1 A. Check your understanding

Exercise. A1

These parts are about Proposition 2.4.1, mainly its part (c).

(a) Where in the proof of (c) is the clause $\varphi(1) = 1$ used?

(b) Let $\zeta\colon \mathbb{Q} \to \mathbb{Q}$ be the zero map. Which clause of Definition 2.4.1 fails? Which of the conclusions (a)–(f) fail for ζ?

(c) Run the proof of (c) with conjugation on $\mathbb{C}$ and the elements $a = 1 + i$, $b = 1 - i$: compute $a - b$, its image, and conclude.

(d) State the idea of the proof of (c) in two sentences.

Solution.

(a) In (b), which (c) cites: $\varphi(a)\varphi(a^{-1}) = \varphi(aa^{-1}) = \varphi(1) = 1$, and $1 \neq 0$, so $\varphi(a) \neq 0$ for $a \neq 0$.

(b) The clause $\zeta(1) = 1$ fails: $\zeta(1) = 0$. Conclusion (a) holds ($\zeta(0) = 0 = -0$). Conclusion (b) fails: $1 \neq 0$ and $\zeta(1) = 0$. Conclusion (c) fails: $\zeta(1) = \zeta(2)$. Conclusion (d) fails: the image $\{0\}$ does not contain 1, so it is not a subfield. Conclusion (f) holds for multiples (both sides are 0) and fails for powers at $n = 0$: $\zeta(a^0) = \zeta(1) = 0$, while $\zeta(a)^0 = 1$. Conclusion (e) fails in its first clause: for a homomorphism $\psi\colon \mathbb{Q} \to M$ (for example $\psi = \mathrm{id}_{\mathbb{Q}}$), $\psi \circ \zeta$ is the zero map, since $\psi(0) = 0$ by (a) for ψ, and the zero map sends 1 to 0, so it is not a homomorphism. The clause about inverses holds vacuously, since ζ is not bijective, and the clause about id_F does not involve ζ.

(c) $a - b = 2i \neq 0$. Its image is $\overline{2i} = -2i \neq 0$, as (b) predicts; so $\overline{a} - \overline{b} = \overline{a - b} \neq 0$, and $\overline{a} \neq \overline{b}$. Indeed $\overline{a} = 1 - i$ and $\overline{b} = 1 + i$.

(d) A non-zero element has an inverse, so its image times another image is $\varphi(1) = 1 \neq 0$; hence a homomorphism sends no non-zero element to 0. Applied to a difference $a - b$, this says that different elements have different images.

2.4.4.2 B. Practice

Exercise. B1: Images of quotients

Let $\varphi\colon F \to L$ be a homomorphism, and let $a, b \in F$ with $b \neq 0$. Prove that $\varphi(a/b) = \varphi(a)/\varphi(b)$. Here is the start of a proof. *Name the move, then complete it.*

"Since $a = (a/b) \cdot b$, applying φ gives $\varphi(a) = \varphi(a/b)\,\varphi(b)$. …"

Solution. The move is **push the equation through** φ: the equation $a = (a/b)b$, which holds since $(ab^{-1})b = a(b^{-1}b) = a$ by (F5), (F6), (F8) and (F7), is carried to L and solved there.

Completion. Since $b \neq 0$, $\varphi(b) \neq 0$ by Proposition 2.4.1 (b), so $\varphi(b)^{-1}$ exists. Multiplying

$\varphi(a) = \varphi(a/b)\varphi(b)$ by $\varphi(b)^{-1}$, with (F5), (F8) and (F7),

$$\varphi(a)\varphi(b)^{-1} = \varphi(a/b)\,\varphi(b)\varphi(b)^{-1} = \varphi(a/b).$$

That is, $\varphi(a/b) = \varphi(a)/\varphi(b)$.

Exercise. B2: The automorphisms of the rational complex numbers

Show that $\mathbb{Q}(i)$ (Example (Rational and Integer Complex Numbers)) has exactly two automorphisms: the identity and conjugation $a + bi \mapsto a - bi$. Here is the first half of a proof. *Name the move, then complete it.*

"Let σ be an automorphism of $\mathbb{Q}(i)$. It fixes every rational: $\sigma(n) = \sigma(n \cdot 1) = n \cdot \sigma(1) = n$ for $n \in \mathbb{N}$ by Proposition 2.4.1 (f), $\sigma(-n) = -\sigma(n) = -n$ by (a), and $\sigma(m/n) = \sigma(m)/\sigma(n) = m/n$ for $m \in \mathbb{Z}$, $n \in \mathbb{Z}_{>0}$, by Exercise (B1: Images of quotients). Moreover $i^2 = -1$. . . ."

Solution. *The move* is **push the equation through** σ, applied to $i^2 = -1$.

Completion. Every rational is m/n with $m \in \mathbb{Z}$, $n \in \mathbb{Z}_{>0}$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.4.4 (b)), so $\sigma(q) = q$ for every $q \in \mathbb{Q}$. Applying σ to $i^2 = -1$,

$$\sigma(i)^2 = \sigma(i^2) = \sigma(-1) = -1 = i^2.$$

By Corollary 2.1.2 in the field $\mathbb{Q}(i)$, $\sigma(i) = i$ or $\sigma(i) = -i$. For $a, b \in \mathbb{Q}$, $\sigma(a + bi) = \sigma(a) + \sigma(b)\sigma(i) = a + b\,\sigma(i)$. So either $\sigma(a + bi) = a + bi$ for all a, b, and σ is the identity, or $\sigma(a + bi) = a - bi$ for all a, b, and σ is conjugation. There are at most these two.

Both occur. The identity is an automorphism (Proposition 2.4.1 (e)). Conjugation sends $a+bi$ to $a+(-b)i \in \mathbb{Q}(i)$, so it maps $\mathbb{Q}(i)$ into itself; it respects sums and products and sends 1 to 1 because conjugation on $\mathbb{C}$ does (Example (Conjugation and Other Automorphisms) (i)) and the operations of $\mathbb{Q}(i)$ are those of $\mathbb{C}$; and it is its own inverse on $\mathbb{Q}(i)$, hence bijective. The two are different maps, since conjugation sends $i \in \mathbb{Q}(i)$ to $-i \neq i$ (Example (Conjugation and Other Automorphisms) (i)). So $\mathbb{Q}(i)$ has exactly two automorphisms.

Exercise. B3

Let F be the field of Exercise (B3: Which Are Fields?) (a): the set $\mathbb{Q}$ with the usual sum and the product $a \odot b := 2ab$. Show that $f\colon F \to \mathbb{Q}$, $f(x) := 2x$, is an isomorphism of fields, so $F \cong \mathbb{Q}$. Compare with the warning of this section.

Solution. The one of F is $1/2$: $a \odot \frac{1}{2} = 2a \cdot \frac{1}{2} = a$. We check the three clauses, with $\mathbb{Q}$'s usual product on the right. Sums: $f(a + b) = 2a + 2b = f(a) + f(b)$, by (F9). Products: $f(a \odot b) = 2(2ab) = (2a)(2b) = f(a)f(b)$, by (F5) and (F6). One: $f(1/2) = 1$. The map is bijective, with inverse $x \mapsto x/2$: $2(x/2) = x$ and $(2x)/2 = x$ (Sets, Relations, and Cardinality, Theorem 1.5.4). So f is an isomorphism.

The warning used the same formula $x \mapsto 2x$ as a map $\mathbb{Q} \to \mathbb{Q}$, with the usual product on **both** sides, and it is not a homomorphism there. Whether a map is a homomorphism

depends on the operations of the source as well as on the formula: on F, the product $\odot$ contains the factor 2 that the formula needs.

Exercise. B4: The fixed field

Let σ be an automorphism of a field F. Show that its **fixed set** $\{a \in F : \sigma(a) = a\}$ is a subfield of F. Find the fixed set of conjugation on $\mathbb{C}$ and of conjugation on $\mathbb{Q}(\sqrt{2})$.

Solution. Write K for the fixed set, and use Proposition 2.3.1 (c). $\sigma(1) = 1$, so $1 \in K$. For $a, b \in K$, $\sigma(a - b) = \sigma(a) - \sigma(b) = a - b$ by Proposition 2.4.1 (a), so $a - b \in K$. If moreover $b \neq 0$, then $\sigma(ab^{-1}) = \sigma(a)\sigma(b)^{-1} = ab^{-1}$ by (b), so $ab^{-1} \in K$. Hence K is a subfield. (Only the homomorphism property was used.)

For conjugation on $\mathbb{C}$, $K = \mathbb{R}$, by Example (Conjugation and Other Automorphisms) (i).

For conjugation on $\mathbb{Q}(\sqrt{2})$, $a - b\sqrt{2} = a + b\sqrt{2}$ holds if and only if $2b\sqrt{2} = 0$, that is, $b = 0$: in the ordered field $\mathbb{R}$, $2 = 2 \cdot 1 > 0$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (h), Number Systems: From \(\nN\) to \(\nR\), Theorem 5.3.3), so $2 \neq 0$, and $\sqrt{2} \neq 0$ (since $0^2 = 0 \neq 2$), and a product of non-zero elements is non-zero (Theorem 2.1.1 (f)). So $K = \{a + 0\sqrt{2} : a \in \mathbb{Q}\} = \mathbb{Q}$.

2.4.4.3 C. Going deeper

Exercise. C1: Isomorphic fields have the same square roots of minus one

(a) Let $F \cong L$. Prove that F has an element x with $x^2 = -1$ if and only if L has one.

(b) Deduce that $\mathbb{Q}(i) \ncong \mathbb{Q}(\sqrt{2})$ and that $\mathbb{C} \ncong \mathbb{R}$.

Solution.

(a) Let $\varphi\colon F \to L$ be an isomorphism. If $x \in F$ has $x^2 = -1$, then by Proposition 2.4.1 (f) and (a),

$$\varphi(x)^2 = \varphi(x^2) = \varphi(-1) = -\varphi(1) = -1,$$

so $\varphi(x) \in L$ is such an element. Conversely, $\varphi^{-1}\colon L \to F$ is an isomorphism (Proposition 2.4.1 (e)), and the same computation carries an element $y \in L$ with $y^2 = -1$ to $\varphi^{-1}(y) \in F$ with $\varphi^{-1}(y)^2 = -1$.

(b) In $\mathbb{Q}(i)$, $i^2 = -1$. In $\mathbb{Q}(\sqrt{2}) \subseteq \mathbb{R}$, every y has $y^2 \geq 0 > -1$, since $\mathbb{R}$ is an ordered field (Number Systems: From \(\nN\) to \(\nR\), Theorem 5.3.3, Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (e), (a)); so $\mathbb{Q}(\sqrt{2})$ has no such element, and by (a), $\mathbb{Q}(i) \ncong \mathbb{Q}(\sqrt{2})$. Likewise $i \in \mathbb{C}$ and no element of $\mathbb{R}$ squares to -1, so $\mathbb{C} \ncong \mathbb{R}$.

The same argument works for any statement written with $+$, $\cdot$, 0, 1 and elements: an isomorphism carries it in both directions. (For $\mathbb{C}$ and $\mathbb{R}$, Example (Pairs of Fields with No Homomorphism) (i) shows more: there is no homomorphism $\mathbb{C} \to \mathbb{R}$ at all.)

Marking scheme. (10 points) - 3: pushes $x^2 = -1$ through φ, with $\varphi(-1) = -1$ justified (by (a) and $\varphi(1) = 1$). - 3: the converse, by either route: through φ^{-1}, with the fact that it is an isomorphism (or a homomorphism) justified (Proposition 2.4.1 (e)); or directly, writing

$y = \varphi(x)$ by surjectivity, getting $\varphi(x^2) = \varphi(-1)$, and concluding $x^2 = -1$ by injectivity. - 2: no element of $\mathbb{Q}(\sqrt{2})$ (or of $\mathbb{R}$) squares to -1, from the order of $\mathbb{R}$. - 2: both conclusions of (b) drawn from (a). Fatal errors (0 for the step): treating $\mathbb{Q}(\sqrt{2}) \ncong \mathbb{Q}(i)$ as evident because the sets "look different"; using $\varphi(-1) = -1$ without (a).

Exercise. C2: Find the error

Here is an argument. Is its conclusion true? Does the argument prove it? If not, name the line that fails and why.

"Every homomorphism $\varphi\colon \mathbb{Q} \to \mathbb{Q}$ is the identity. (1) By definition, $\varphi(1) = 1$, and φ respects sums. (2) For $x \in \mathbb{Q}$, $\varphi(x) = \varphi(x \cdot 1) = x\,\varphi(1)$. (3) Hence $\varphi(x) = x \cdot 1 = x$."

Solution. The conclusion is **true**: the repair below proves it, and Section 2.5 proves it again by another route. The argument does not prove it.

Line (2) fails, at the second equality. "$\varphi(x \cdot 1) = x\,\varphi(1)$" pulls the factor x out of φ, as if φ were compatible with multiplication by a scalar x; that is not a clause of Definition 2.4.1, and nothing proved so far supplies it. So the line cites a general "fact" that is not available, entry (N9) of Logic and Proof, Example (What is not allowed). The clause that is available, multiplicativity, gives only $\varphi(x \cdot 1) = \varphi(x)\varphi(1) = \varphi(x)$, which proves nothing.

Counterexample to the step (in its general form, "a homomorphism satisfies $\varphi(x \cdot 1) = x\,\varphi(1)$"): conjugation on $\mathbb{C}$ has $\overline{i \cdot 1} = -i$, while $i \cdot \overline{1} = i$.

Repair. For $n \in \mathbb{N}$, $\varphi(n \cdot 1) = n \cdot \varphi(1) = n \cdot 1$ by Proposition 2.4.1 (f), and $n \cdot 1 = n$ in $\mathbb{Q}$; so $\varphi(n) = n$. For $n \in \mathbb{N}$, $\varphi(-n) = -\varphi(n) = -n$ by (a), so $\varphi(m) = m$ for every $m \in \mathbb{Z}$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.2.3 (c)). Every rational is m/n with $m \in \mathbb{Z}$, $n \in \mathbb{Z}_{>0}$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.4.4 (b)), and $\varphi(m/n) = \varphi(m)/\varphi(n) = m/n$ by Exercise (B1: Images of quotients). So $\varphi = \mathrm{id}_{\mathbb{Q}}$.